



AI-Powered Credit Risk Intelligence Platform

PREDICTIVE ANALYTICS, EXPLAINABLE AI, AND GENERATIVE AI FOR
CREDIT RISK ASSESSMENT

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Problem Statement

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- Financial institutions face significant challenges in assessing the creditworthiness of loan applicants. Incorrect lending decisions can result in financial losses due to loan defaults, while overly strict policies may reject eligible customers.

Challenges

- Large volume of loan applications
- Complex customer financial profiles
- High risk of loan defaults
- Need for transparent and explainable decisions

Proposed Solution

- Develop an AI-powered platform that predicts loan default risk, explains prediction outcomes, and enables natural language interaction with credit data through a conversational chatbot.

Dataset Overview

Home Credit Default Risk Dataset

- The project utilizes the Home Credit Default Risk dataset containing information about loan applicants, demographic attributes, financial history, and repayment behavior.

Dataset Statistics

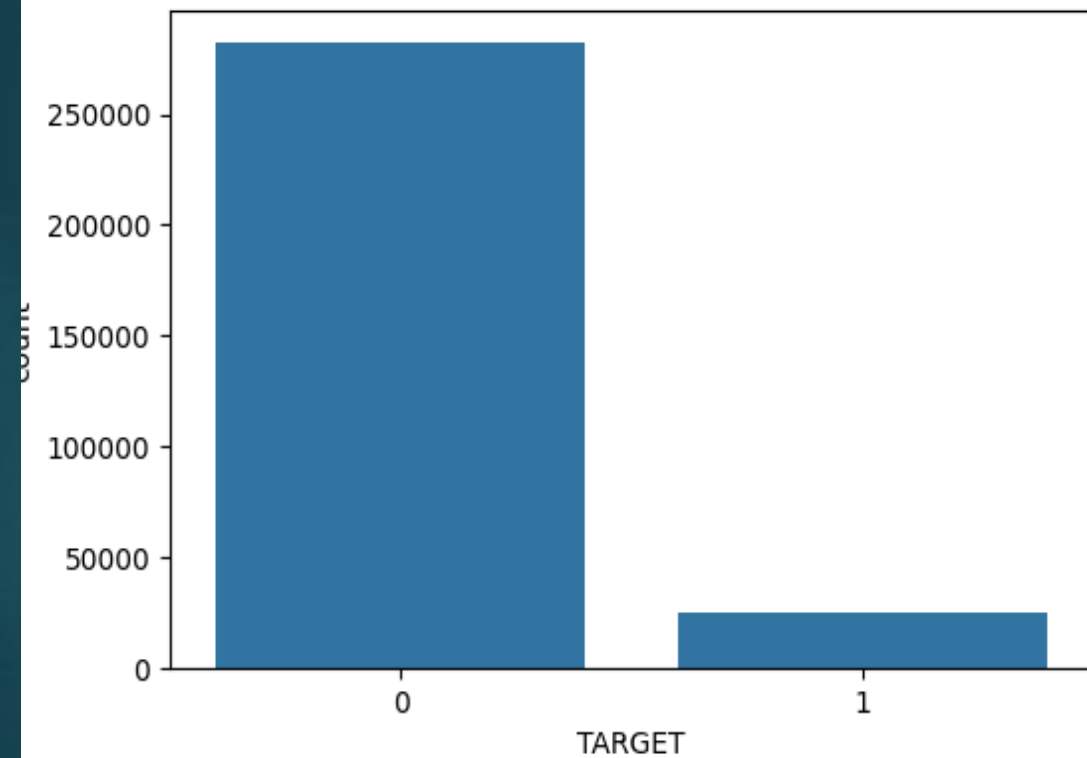
- Total Records: 307,511
- Total Features: 122
- Target Variable: TARGET
 - 0 = Non-Default
 - 1 = Default

Key Features

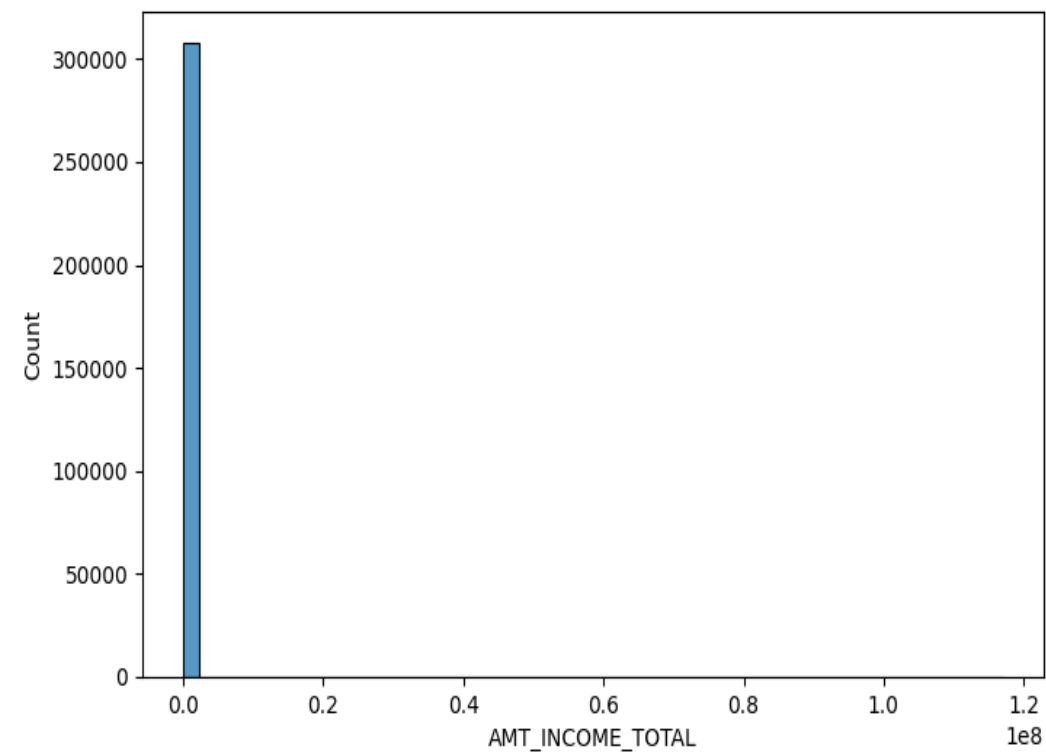
- Applicant Income
- Credit Amount
- Loan Annuity
- Age
- Employment Duration
- External Risk Scores



Loan Default Distribution



Income Distribution



Exploratory Data Analysis

Exploratory Data Analysis (EDA)

- EDA was performed to understand applicant characteristics, identify data quality issues, and discover relationships between variables affecting loan repayment behavior.

Key Findings

- Dataset contains missing values and outliers.
- Income levels vary significantly across applicants.
- External risk scores strongly influence default probability.
- The target variable is highly imbalanced.

Model Development

Machine Learning Model

- A LightGBM classifier was selected to predict the probability of loan default.

Why LightGBM?

- Efficient handling of large datasets
- High performance on structured/tabular data
- Fast training and inference
- Ability to handle complex feature interactions

Model Pipeline

- Data Cleaning
- Missing Value Imputation
- Label Encoding
- Feature Selection
- Model Training
- Performance Evaluation

Model Performance

Model Evaluation Results

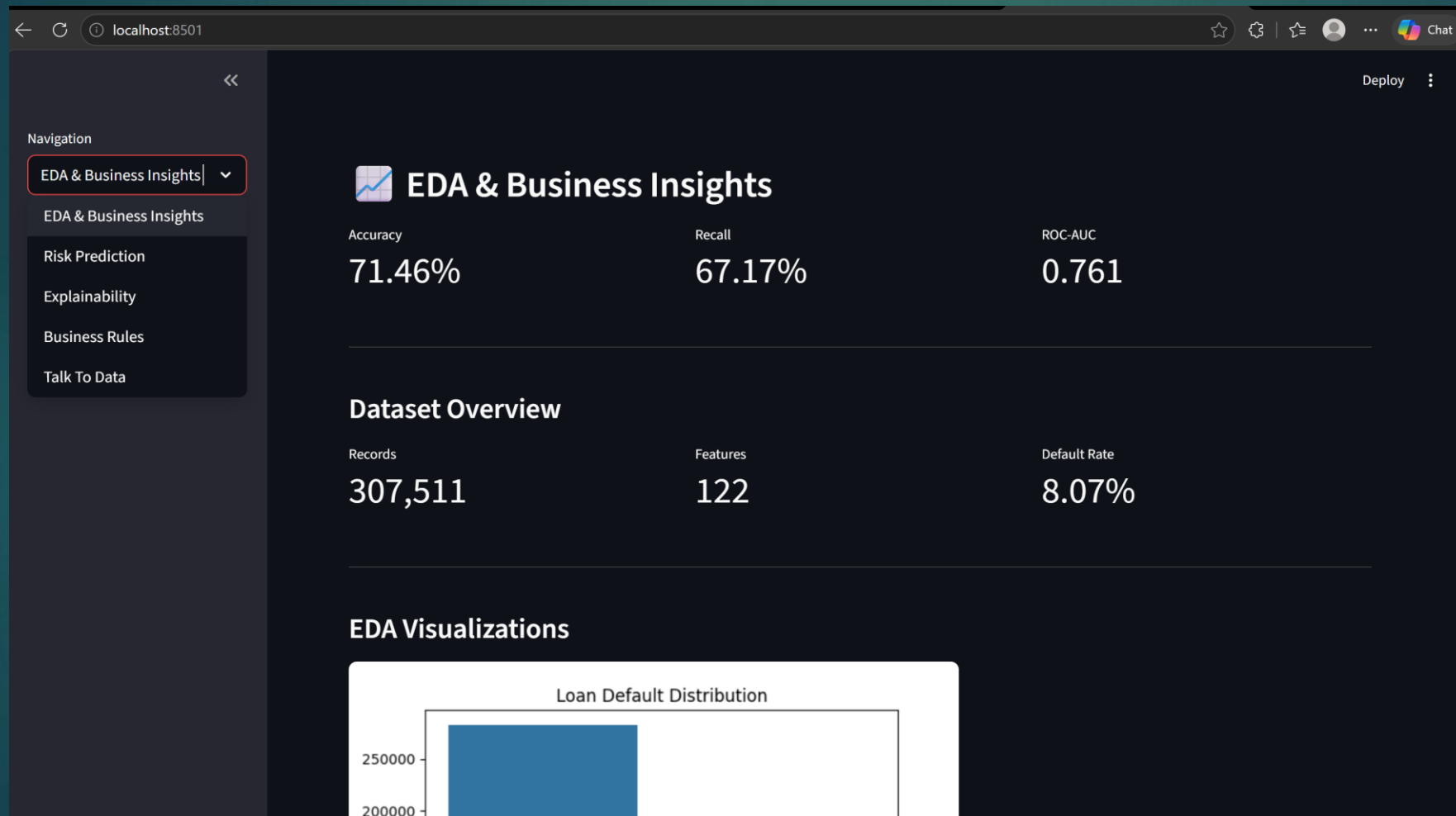
The trained LightGBM model was evaluated using multiple classification metrics to ensure reliable risk prediction.

Key Insight

The model achieved a strong ROC-AUC score, indicating good discrimination between default and non-default applicants.

Metric	Score
Accuracy	71.46%
Precision	17.32%
Recall	67.17%
F1 Score	27.53%
ROC-AUC	76.13%

Dashboard



Explainable AI using SHAP

SHAP Explainability

- Financial decision-making requires transparency. SHAP (SHapley Additive Explanations) was used to explain how individual features contribute to model predictions.

Benefits

- Improves model interpretability
- Supports regulatory compliance
- Builds stakeholder trust

Most Influential Features

- EXT_SOURCE_1
- EXT_SOURCE_2
- EXT_SOURCE_3
- DAYS_BIRTH
- AMT_CREDIT

Navigation

Explainability



SHAP Explainability

SHAP (SHapley Additive Explanations) helps explain which features influence credit risk predictions the most.



Deploy



Credit Risk Prediction Dashboard

Interactive Risk Assessment

- A Streamlit-based dashboard was developed to allow users to enter applicant information and receive real-time risk predictions.

User Inputs

- Income
- Credit Amount
- Loan Annuity
- Age
- Gender
- Education
- Property Ownership

Output

- Default Probability
- Risk Category
- Risk Interpretation

Navigation

Risk Prediction

Deploy

Credit Risk Prediction

Applicant Income

150000.00

-

+

Credit Amount

300000.00

-

+

Loan Annuity

25000.00

-

+

Goods Price

200000.00

-

+

Age

30

-

+

Years Employed

5

-

+

Gender

M

Own Car

Y

Own Property

Y

Education

Higher education

Family Status

Married

External Score 1

0.50

Navigation

Risk Prediction

Deploy

Predict Risk

External Score 2

0.50

External Score 3

0.50

	EXT_SOURCE_1	EXT_SOURCE_2	EXT_SOURCE_3	AMT_INCOME_TOTAL	AMT_CREDIT
0	0.5	0.5	0.5	150000	300000

Prediction Result

Default Probability

32.03%

LOW RISK (32.03%)

Gemini-Powered Talk-to-Data Chatbot

Generative AI Integration

- The platform integrates Google's Gemini model to enable natural language interaction with the credit risk database.

Capabilities

- Converts user questions into SQL queries
- Retrieves insights directly from the database
- Simplifies data exploration for business users

Example Query

- "What is the average income of defaulters?"
- The chatbot automatically generates SQL and returns the corresponding result.

Navigation

Talk To Data



Deploy 

Talk To Data

Ask a business question

What is the average income of defaulters?

Generate Answer

Generated SQL

```
SELECT AVG(AMT_INCOME_TOTAL) AS average_income
FROM applications
WHERE TARGET = 1;
```



Query Result

	average_income
0	165611.7609

Docker Deployment & Conclusion

Docker Deployment

- The application was containerized using Docker to ensure portability, scalability, and consistent execution across environments.

Benefits

- Easy deployment
- Environment consistency
- Simplified dependency management

Conclusion

- Developed an end-to-end Credit Risk Intelligence Platform.
- Integrated Machine Learning, Explainable AI, and Generative AI.
- Enabled automated risk assessment and data-driven decision-making.
- Delivered a deployable and scalable solution using Docker.



Thank You